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Place Effects and Geographic Inequality in Health at Birth†

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This paper uses birth records and mothers who move to quantify the absolute and relative importance of birth location for early-life health. Using a model that includes mother and location fixed effects, we find that moving from a below- to an above-median-birth-weight location leads to important improvements in child birth weight, with comparable magnitudes to policies targeting maternal health. Place effects are larger for longer-distance moves and more influential for children of non-college-educated mothers. We find that pollution is the strongest predictor of place effects on infant health. (*JEL* I12, I14, J13, J16, Q51, Q53)

Although the United States strives to be a land of equal opportunity, geographic disparities in the outcomes of children appear as early as birth. For example, children born in the Midwestern cities of Detroit and Cleveland are more than twice as likely to be born with low birth weight (below 2,500 grams) compared to those born in coastal cities like San Diego and Seattle (Kids Count Data Center 2021). Because of the link between birth weight and later-life success, this raises the potential concern that disparities in the quality of a child's birth location may translate into gaps in economic success and upward mobility.[1] However, it remains unclear the extent to which these disparities in early life reflect the causal impact of place rather than the nonrandom sorting of families.

This paper provides new evidence on the role of place in early-life outcomes. Using birth records from California that span three decades, we follow "mover" mothers that relocate across neighborhoods and compare the birth weights of children born postmove relative to previous births to identify causal place effects. Our within-mother estimates capture the total effect of place, including the impacts of hard-to-measure characteristics such as the degree of social capital and more salient features such as pollution and climate. We use the estimated place effects to quantify the impact of moving to higher-quality neighborhoods and decompose spatial

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[1] See Black, Devereux, and Salvanes (2007); Chetty et al. (2014); and Bharadwaj, Lundborg, and Rooth (2018).

gaps in early-life health into a causal, place-based component and a selection, family-based component.

Our first main result is that the quality of an infant's birth location (as measured by the outcomes of other infants born there) is influential for birth weight. In our primary specification, mothers who move from a below- to an above-median zip code experience an 18-gram improvement in their own child's birth weight. This compares favorably with the impact of policies that target pregnant women's health, such as providing access to Medicaid (East et al. 2021) or reducing smoking (Permutt and Hebel 1989). Place effects are symmetric for increases and decreases in the quality of location, suggesting that mothers are sensitive to both the health benefits and the health costs of locations. We find positive effects throughout the birth weight distribution, which indicates that place is influential even for relatively healthy pregnancies.

The second main finding is that place effects are larger for longer-distance moves. We establish this by using variation in the distance of moves across zip codes, as well as by reestimating place effects at the city and county levels—higher levels of geographic aggregation that mechanically identify the average effects of larger distance moves. A key finding in this analysis is that county-level moves have the largest impact: Moving from a below- to an above-median county increases birth weight by 26 grams, an effect that is 44 percent larger than our estimate for analogous zip code–level moves. To explain these results, we provide evidence that county-level moves lead to larger changes in observable environmental inputs that shape early child health.

Our third finding comes from our decomposition analysis which shows that our zip code–level place effect estimates account for 17 percent of the 109-gram difference in birth weight between above- and below-median areas, implying that the remaining 83 percent of the variation is due to sorting on mother-specific factors. This indicates that equalizing location-specific amenities could eliminate roughly one-sixth of the spatial gap in infant health. We find similar shares when we decompose the gap between the top and bottom quartiles or deciles. When we use city- and county-level estimates, our decomposition results still show that place can explain less than half of the gaps between above- and below-median areas.

Fourth, not all groups are equally affected by place; we find much larger effects of moving for mothers without a college degree. Moving from a below-median to an above-median zip code improves child birth weight for non-college-educated mothers by over three times as much as for college-educated mothers (e.g., 22 grams versus 7 grams). In turn, place explains significantly more of the spatial gap in outcomes for non-college-educated mothers (e.g., 20 versus 6 percent). These results add to an emerging body of evidence that contextual factors such as pollution may have stronger impacts on disadvantaged populations (e.g., Currie and Walker 2011).

To shed light on the mechanisms behind the impacts of place, we study the area-level correlates of our estimates of causal place effects. We find that pollution—particularly, the local level of ozone—is the strongest correlate of place effects. This suggests that one of the primary ways that place affects fetal development is by shaping the level of maternal exposure to pollutants. Correlations between place effects and the supply of prenatal care and average maternal education are smaller but meaningful, suggesting potential roles for improved access to health care and the preferences of local residents.

These results rely on the key assumption that changes in the family-based determinants of birth weight do not correlate systematically with the improvement in the quality of a mother's location. We address potential threats to this assumption in three ways. First, we use an event-study approach to rule out pre-trends and show that changes in birth weight only appear for the children born after a move. Second, we provide suggestive evidence on the role of confounding changes by estimating the impact of a move on *predicted* birth weight, which we construct from a large number of time-varying maternal and paternal characteristics, including proxies for economic status (some of which may be influenced by place). Our results suggest that changes in these characteristics explain less than 3 percent of the place effect. Third, we show that our results are similar if we instrument for the quality of a mother's destination using the location choices of *other* mover mothers in her origin or focus on moves among families who are attached to the military, who may have less control over the timing or locations of moves (e.g., Lyle 2006; Carter and Wozniak 2018).

Our study makes three contributions to the literature. First, we quantify the *total* role of place-based factors in infant health. In contrast, prior research focuses on estimating the impacts of specific contextual characteristics on early-life health, such as the effects of exposure to pollution or higher temperatures on infant health at birth or child mortality (Chay and Greenstone 2003; Currie and Neidell 2005; Currie, Neidell, and Schmieder 2009; Deschenes, Greenstone, and Guryan 2009; Currie, Greenstone, and Moretti 2011; Currie and Walker 2011; Currie et al. 2015; Knittel, Miller, and Sanders, 2016; Alexander and Schwandt, 2022; Heft-Neal et al. 2020; Hansen-Lewis and Marcus 2022).[2,3] Moreover, our data are unique in allowing us to follow the individual mothers over time rather than using a county-level (or similar geographic unit) panel with aggregate health outcomes. This allows us to rule out changes in the sample composition as a possible source of confounding variation for the effect of place.

Second, our results complement previous studies of the causal impact of place on health-related behaviors and mortality outcomes for adults (Finkelstein, Gentzkow, and Williams 2016, 2021; Deryugina and Molitor 2020; Hinnosaar and Liu 2022). Relative to prior estimates, our findings suggest that the share of geographic disparities in early-life health explained by place-based factors is smaller than the share for health spending (Finkelstein, Gentzkow, and Williams 2016) and alcohol consumption (Hinnosaar and Liu 2022) but at least as large as the share for over-65 mortality (Finkelstein, Gentzkow, and Williams 2021). This is consistent with a hypothesis in which location has a larger influence over inputs to health capital

[2] In a similar vein, there is also a literature outside of economics that documents associations between features of the local environment and infant health (e.g., Morenoff 2003; O'Campo et al. 2008; Kramer et al. 2012; Witt et al. 2015; Himmelstein and Desmond 2021).

[3] Our analysis of determinants of infant outcomes also complements research estimating the effects of the role of family background in early-life health, such as the effect on birth outcomes of a mother's highest level of completed education (McCrary and Royer 2011; Currie and Moretti 2003), nutrition intake (Hoynes, Page, and Stevens 2011; Almond, Hoynes, and Schanzenbach 2011; Rossin-Slater 2013), insurance coverage (Currie and Gruber 1996; East et al. 2021; Miller and Wherry 2022), and financial resources (Dehejia and Lleras-Muney 2004; Lindo 2011; Hoynes, Miller, and Simon 2015).

relative to the stock of health capital. Methodologically, our analysis is most closely related to Finkelstein, Gentzkow, and Williams (2016), who also use a decomposition approach to study effects on health care spending. One advantage of our study is that the data allow us to uniquely test for and rule out potential confounds related to changes in family circumstances or maternal income that occur after a move. This provides additional support for using a mover-based design in this context and may extend to other settings as well.

Finally, we add to a growing literature studying children and neighborhood effects. Previous work has shown that living in a better neighborhood during childhood and adolescence leads to significant improvements in later-life labor market activity, criminal behavior, and education (Damm and Dustmann 2014; Chetty, Hendren, and Katz 2016; Chyn 2018; Chetty and Hendren 2018; Chetty et al. 2020a, b; Chyn, Collinson, and Sandler 2025; Laliberte 2021; Baran, Chyn, and Stuart 2024). To our knowledge, we are the first to estimate causal place effects for outcomes at birth. This fills a gap in earlier findings, which measure the impacts of place on other outcomes during older ages. The projected effects on earnings from our estimates suggest that moving to a higher-quality neighborhood could have a small impact on later-life outcomes by improving health at birth. An implication of this finding is that the previously documented impacts of neighborhoods on long-run outcomes are likely to primarily operate through postbirth exposure rather than through in utero development. Hence, prior evidence that neighborhood exposure effects taper at younger ages likely also extends to the critical prenatal period (Deutscher 2020; Chetty et al. 2020a, b; Chetty and Hendren 2018).

I. Empirical Model

Our analysis consists of two main exercises. First, we use a mover design to estimate the causal effect of place on early-life health. As detailed in Section III, we primarily define place as a zip code, but we also consider city- and county-level definitions in supplemental analyses. As a second exercise, we use the estimated place effects to decompose the difference in infant outcomes across areas into contextual (place-specific) factors *or* family characteristics. The model of place effects and our subsequent decomposition analysis are similar to prior research on the determinants of medical spending among the elderly (Finkelstein, Gentzkow, and Williams 2016).[4]

A. *Causal Place Effects*

We estimate the impact of birth location on early-life health using the following model for child outcomes:

$$y^{c}_{mjkt} = \alpha_m + \gamma_j + \theta_k + \tau_t + \rho_{r(m,k)} + \mathbf{x}'_{mk}\boldsymbol{\beta} + \epsilon^{c}_{mjkt}, \tag{1}$$

[4] Other studies use similar approaches to examine non-health-related behavior, such as consumer financial health (Keys, Mahoney, and Yang 2020), voting (Cantoni and Pons 2022), and workers' earnings (Card, Rothstein, and Yi 2021).

where y^c_{mjkt} is the birth outcome for child c (e.g., birth weight) for the kth birth of mother m who lives in geographic area j in year t. The terms α_m, γ_j, θ_k, and τ_t represent mother (i.e., family), location, birth-order, and calendar year fixed effects, respectively. For mothers who move across areas, $r(m,k) = k - k^*$ is an index that tracks the birth order relative to the first postmove birth k^*. For example, $r(m,k) = 0$ if k is the first birth that occurs in a new mother's new location, and $r(m,k) = -1$ if k is the last birth that preceded a move.[5] The term $\rho_{r(m,k)}$ is a fixed effect for this index. For mothers who never move, we assume $r(m,k) = 0$. We also include fixed effects for the sex of the child and for five-year bins of maternal age, $\mathbf{x}_{mk}$. Finally, ϵ^c_{mjkt} is an error term that we assume is conditionally mean zero: $E\left[\epsilon^c_{mjkt} \mid m,j,k,t,x_{mk}\right] = 0$.

The key parameters of interest are the location fixed effects, γ_j. These are identified only if the estimation sample includes "mover" mothers who have multiple births and relocate between births. Otherwise, if all mothers gave birth in the same location or only had one birth, we could not separately identify the impact of location (γ_j) from the unobserved heterogeneity across families (α_m). As discussed in Section III, we include both mover and nonmover mothers in our sample to enable us to identify the birth-order fixed effects (which are collinear with $r(m,k)$ for movers), but our qualitative conclusions remain the same if we drop nonmovers.[6]

In order to interpret γ_j as the *causal* impact of place, we require that changes in unobserved, family-based determinants of early-life outcomes are not correlated with the difference in the average outcomes between the destination and origin chosen by a mother. For example, this assumption would be violated if mothers who receive negative shocks (that are correlated with child outcomes) respond by moving to areas that have worse place effects. This would lead us to attribute some of the mother-specific adverse shock to the effect of moving and, thus, overstate the place effect. We discuss the testable implications of this assumption in Section II.

B. *Decomposition*

We use the model from equation (1) to guide our decomposition of geographic disparities in infant outcomes. Formally, let $\bar{y}_j$ denote the expectation of y^c_{mjkt} across mothers living in location j. Also, let $\bar{y}^m_j$ denote the expectation of the part of child outcomes that is only attributable to family (mother) characteristics. This includes the influence of all fixed traits for a given mother (i.e., α_m), as well as predictable changes in outcomes that occur across children within a mother, such as from increasing birth order (i.e., $\theta_k + \tau_t + \rho_{r(m,k)} + \mathbf{x}'_{mk}\beta$ in our model).[7] Using this notation, equation (1) implies that $\bar{y}_j = \bar{y}^m_j + \gamma_j$.

[5] Because we observe few mothers either three births before a move $(r(m,k) = -3)$ or three births after a move $(r(m,k) = 2)$, we group $r(m,k) = -3$ with $r(m,k) = -2$ and $r(m,k) = 2$ with $r(m,k) = 1$.

[6] While fertility may be an endogenous response to moves, we find that a mother's total number of children is uncorrelated with her location decision (see Section VIB).

[7] The inclusion of these predictable changes in outcomes in the individual component of the decomposition is consistent with prior work (Finkelstein, Gentzkow, and Williams 2016).

Applying these definitions, the difference in average child outcomes between locations j and j' can be written as the sum of two components:

$$\text{(2)} \qquad \bar{y}_j - \bar{y}_{j'} \;=\; \left(\gamma_j - \gamma_{j'}\right) + \left(\bar{y}_j^m - \bar{y}_{j'}^{m'}\right).$$

The first term in equation (2) is a place-specific component given by the difference in location fixed effects, $\gamma_j - \gamma_{j'}$. This is the causal effect of moving from location j to j'. The second term is a family-specific component given by the difference in average maternal characteristics, $\bar{y}_j^m - \bar{y}_{j'}^m$. Rearranging terms, the share of the difference in outcomes between locations j and j' that is attributable to place is

$$\text{(3)} \qquad S_{place}\left(j,j'\right) \;=\; \frac{\gamma_j - \gamma_{j'}}{\bar{y}_j - \bar{y}_{j'}}.$$

Analogously, the share attributable to family (mother) factors is

$$\text{(4)} \qquad S_{mom}\left(j,j'\right) \;=\; \frac{\bar{y}_j^m - \bar{y}_{j'}^{m'}}{\bar{y}_j - \bar{y}_{j'}}.$$

By construction, the sum of these place- and family-specific shares, $S_{place}\left(j,j'\right)$ and $S_{mom}\left(j,j'\right)$, is equal to 1.

Empirically, we apply these formulas to decompose average outcomes in *groups* of locations, R and R', such as the top and bottom 50 percent of locations. We define the shares for these groups as $S_{place}\left(R,R'\right)$ and $S_{mom}\left(R,R'\right)$, which we compute by replacing the j- and j'-level inputs in the equations above with averages within R and R', respectively. We obtain standard errors for our estimates as the standard deviation of the quantity of interest across 50 bootstrapped samples.

As an alternative to the above additive decomposition, we also decompose the variance in birth weight across places. Here, we study the share of cross-location variance in birth weight that would be eliminated in counterfactuals where either the average maternal characteristics or place (zip code) effects were equalized across locations. These quantities are

$$S_{mom}^{var} \;=\; 1 - \frac{\operatorname{var}\left[\gamma_j\right]}{\operatorname{var}\left[\bar{y}_j\right]},$$

$$S_{place}^{var} \;=\; 1 - \frac{\operatorname{var}\left[\bar{y}_j^m\right]}{\operatorname{var}\left[\bar{y}_j\right]}.$$

Note that the sum of S_{mom}^{var} and S_{place}^{var} will not equal one as long as $\operatorname{cov}\left[\bar{y}_j^m, \gamma_j\right]$ is nonzero. We use a split-sample approach to estimate this covariance and the variances in the above equations, as well as to generate bootstrapped standard errors for the estimated shares.[8]

[8] As in Finkelstein, Gentzkow, and Williams (2016), we randomly assign movers within each origin-destination pair and nonmovers within each zip code to two approximately equal-sized subsamples, and estimate equation (1) on each subsample. We estimate the variance of γ_j and $\bar{y}_j^m$ as the covariance between the estimates of γ_j and $\bar{y}_j^m$ from

II. Event Study and Tests of Identification

We implement three tests to study potential violations of the identifying assumption required for causal interpretation of our place effect estimates. Although we cannot directly rule out all violations of this assumption (i.e., that the improvement in the quality of a mother's location after a move is uncorrelated with family-based determinants of infant health), the exercises that we consider narrow the scope for potential concerns. This section previews these tests, while Sections IVA and VI provide results and discussion.

As one of our main tests, we use an event-study approach to test for differences in within-mother changes in child outcomes between mothers who move to higher- and lower-quality destinations *prior* to a move. This addresses potential concerns about differential pre-trends, which could bias our results. To implement this test, we rely on the following event-study model:

$$y^c_{mjkt} = \alpha_m + \sum_{r=-2}^{1} \theta_{r(m,k)} \hat{\delta}_m + \omega_k + \nu_t + \rho_{r(m,k)} + \mathbf{x}'_{mk}\boldsymbol{\eta} + \varepsilon^c_{mjkt}, \tag{5}$$

where $\hat{\delta}_m$ is the difference in average birth weight between the mother's origin $o(m)$ and destination $d(m)$, which we use as a proxy for the change in the quality of a mother's destination.[9] The main parameters of interest are the relative birth coefficients $\theta_{r(m,k)}$. These parameters represent the change in the outcome y^c_{mjkt} in the years around the move scaled by the local area differences in birth weights $\hat{\delta}_m$. For instance, a positive value of $\theta_{r(m,k)}$ implies that moving to a location that has better birth outcomes is associated with improvements in the birth outcomes of one's own children in relative year $r(m,k)$. We estimate the event study using all mothers (although the results are the same when we restrict the estimation sample to mover mothers). We include the same control variables as in our model in equation (1) (i.e., birth order, year, birth order relative to move, and sex of child, binned maternal age). For inference, we cluster standard errors by one's origin location (e.g., zip code, at baseline).

If move-induced changes in place characteristics cause changes in child outcomes, then we should observe the following pattern. The relative quality of one's destination should not be predictive of the birth outcomes of one's children *before* a move but should be predictive of outcomes *after* a move. This implies that the estimate of $\theta_{r(m,k)}$ should be statistically indistinguishable from zero for any birth preceding a move (i.e., $r(m,k) < 0$) and nonzero for all births following a move. The magnitude of the discontinuity in the level of $\theta_{r(m,k)}$ after a move measures the extent to which place-specific factors influence child outcomes.[10]

the two subsamples. The estimated correlation between γ_j and $\bar{y}^m_j$ is based on the estimated variance of γ_j and $\bar{y}^m_j$ and the covariance of γ_j and $\bar{y}^m_j$, which we compute as the average of the covariances between the estimates of γ_j from one subsample and $\bar{y}^m_j$ from the other subsample.

[9] We measure $\hat{\delta}_m = \hat{\bar{y}}_{d(m)} - \hat{\bar{y}}_{o(m)}$ using leave-one-out means for our estimation sample that omit the outcomes of mother m.

[10] Due to data limitations, we do not observe the precise timing of moves and cannot pinpoint when the first birth occurs after a move. Based on the average spacing between births before and after a move observed in birth records, a mother could be expected to have resided in a destination zip code for up to 4.5 years.

Next, we conduct two additional types of tests to address potential sources of endogeneity. The first of these tests is motivated by the possible concern that the timing of a move may be correlated with shocks to maternal and household outcomes that affect child health. To address this, we study the extent to which changes in time-varying observable characteristics of mothers can predict any post-move changes in birth weight. This exercise provides evidence for the potential role of coinciding *shocks* with a move. On the other hand, this test may also capture how maternal behaviors might respond to a move, which could be a *mechanism* for place effects.[11] We consider both of these interpretations in our discussion.

The second type of test addresses the potential concern that the choice of destination may be endogenous. Specifically, we take advantage of two quasi-random sources of changes in location quality ("shocks") to assess the sensitivity of our conclusions. As one shock to location quality, we use an instrumental variable (IV) approach to isolate variation arising from systematic patterns of improvements in location quality among individuals who move from a given origin. The power of the instrument comes from a regression-to-the-mean-type logic: Individuals that begin in worse origins are more likely to move to a relatively better destination, and individuals that begin in better origins are more likely to move to a relatively worse destination. This idea is similar to the IV approaches in Chetty and Hendren (2018) and Abaluck et al. (2021). As another shock to location quality, we focus on moves among families attached to the military, whose origin and destination may be more likely to be determined by army staffing needs than individual preferences (e.g., Lyle 2006; Carter and Wozniak 2018).

III. Data

Our primary data source is confidential individual birth records from California from 1989 to 2017. The data are compiled from forms completed at birth and contain approximately 15.6 million records. For each birth, the data include infant outcomes such as birth weight and length of gestation. In addition, the records contain information on the identity of the mother, her residential address, and demographic characteristics such as her place of birth, race, date of birth, and educational background and proxies for her economic status (e.g., type of insurance).[12]

Our main outcome of interest is birth weight (measured in grams), a key measure of early-life health that has been widely studied (Almond and Currie 2011). Previous research links birth weight to a range of long-run outcomes, including education, adult health, and earnings (Black, Devereux, and Salvanes 2007; Royer 2009; Figlio et al. 2014; Bharadwaj, Eberhard, and Neilson 2018). We focus on birth weight because other outcomes, such as very low birth weight status, are rare, particularly at more granular levels of geography.

[11] For example, mothers who move to higher-quality locations (as measured by child health outcomes) could obtain a new job that provides access to health insurance. This type of place-based mechanism would be reflected in our estimated place effects.

[12] We use this information to construct a unique identifier for each mother based on first name, maiden name, date of birth, and place of birth.

To study the role of place, we use residential zip code (ZIP-5) as the primary geographic unit. This unit was created by the US Postal Service and represents small geographic areas (typically with populations less than 10,000). Using this fine level of geography allows us to capture nuanced differences across neighborhoods, such as the diffusion of knowledge about public programs (Chetty, Friedman, and Saez 2013). Our estimation sample includes 1,690 zip codes. In Section IVF, we also explore place effects using the cities and counties as the geographic unit of interest. There are 1,010 cities and 58 counties covered in the California birth records.

A. *Sample*

Two main restrictions define the sample of mothers for our analysis. First, we focus on mothers who are California residents at the time of childbirth and who we observe having two to four births during the period covered by our records. The restriction to mothers with multiple children is necessary so that we have multiple observations with which to infer the mobility of mothers over time. We define a nonmover as a mother who is observed in the same geographic location (e.g., zip code) for all her births. A mover is a mother who changes locations exactly once; we drop mothers with multiple moves.[13] Second, to reduce noise in our estimates, the sample is restricted to mothers who live in an area that contains at least 25 movers.

Table 1 reports summary statistics for all births and for our estimation sample. Columns 1 and 2 show that the births in our estimation sample have broadly similar birth weight and demographic characteristics relative to all births in California.[14] Our estimation sample includes roughly 3.3 million mothers with a total of 7.4 million births. Among this sample, 53 and 47 percent of births are to nonmover and (zip code) mover mothers, respectively. Columns 3 and 4 show that movers and nonmovers are substantively similar in terms of infant birth weight and ethnoracial demographics, although movers have less education and are younger.

Supplemental Appendix Figure A2 provides a map of birth weight at the zip code level, our primary geography of interest. The average birth weight in the median zip code is 3,349 grams, and the standard deviation across zip codes is 59 grams. We also find a similar distribution when we aggregate the birth records to the county level: The mean birth weight in the median county is 3,361 grams, and the standard deviation across counties is 55 grams.

How do these statistics for California compare more broadly? Based on birth records from the National Center for Health Statistics (NCHS) in 2004, which is roughly the median year of our data, the mean birth weight in the United States is 3,291 grams, which is 1 percent less than what we find for California (National Center for Health Statistics 2004).[15] The average within-state standard deviation of

[13] Mothers with more than four births account for 3 percent of births in our sample. Mothers with multiple moves are 13 percent of the sample. The results in Supplemental Appendix Table A12 show that our estimates are similar if we include mothers with multiple moves.

[14] Supplemental Appendix Figure A1 shows that the distributions of birth weight for all births in California and the births in our estimation sample are not meaningfully different.

[15] We calculate these statistics using the 2004 NCHS National Vital Statistics System birth files, which include the universe of US birth records and contain identifiers for counties with populations over 100,000. The 2004 records are the last year to include county identifiers.

Table 1—Summary Statistics

	All	Estimation sample		
		All	Nonmovers	Zip code movers
Infant birth weight (grams)	3,332.766	3,338.882	3,321.158	3,358.838
	(577.5)	(573.7)	(591.9)	(551.8)
Black mother	0.066	0.053	0.040	0.069
	(0.248)	(0.225)	(0.195)	(0.253)
White, non-Hispanic mother	0.792	0.804	0.811	0.795
	(0.406)	(0.397)	(0.391)	(0.404)
Hispanic mother	0.483	0.456	0.447	0.466
	(0.500)	(0.498)	(0.497)	(0.499)
Asian mother	0.087	0.091	0.099	0.081
	(0.282)	(0.287)	(0.298)	(0.274)
Mother has high school degree	0.721	0.771	0.780	0.762
	(0.448)	(0.420)	(0.414)	(0.426)
Mother has college degree	0.226	0.274	0.311	0.233
	(0.418)	(0.446)	(0.463)	(0.423)
Maternal age	27.958	28.144	28.761	27.450
	(6.260)	(6.007)	(6.002)	(5.936)
Number of births	2.208	2.421	2.357	2.494
	(1.141)	(0.630)	(0.594)	(0.661)
Observations	15,139,133	7,412,595	3,925,885	3,486,710

Notes: This table presents summary statistics based on birth records from California (1989–2017). Column 1 provides statistics for all births. Columns 2–4 provide statistics for the estimation sample that we use for our main analysis of zip code movers.

birth weight across counties nationally is 43 grams, which is 12 grams (22 percent) lower than our estimate for California.[16] Overall, we interpret these small differences as reassuring in that our setting is not particularly unusual relative to other states.

Finally, we provide descriptive statistics on the types of moves (using all cross-zip-code moves). Supplemental Appendix Table A1 shows that the average and median distances moved across zip codes are 31.2 and 8.8 miles, respectively. We also find that roughly 28 percent of these moves cross county boundaries. Movers' destination zip codes receive an average of 7,982 mothers at some point after their first birth. Figure 1 further characterizes moves in terms of the change in location quality. As in the event-study model, we measure the improvement in location quality using the difference in average birth weight between a mother's destination and origin ($\hat{\delta}_m$). The distribution of $\hat{\delta}_m$ is centered around zero and is roughly symmetric, indicating that mothers are equally likely to move to a better or worse location (based on average birth weight). The standard deviation is 25 grams, which is equivalent to the change experienced from moving from the median zip code to the sixty-fifth percentile zip code.

[16] Note that the average within-state standard deviation (43 grams) is roughly three-quarters of the unconditional standard deviation across counties, suggesting that the majority of across-county variation is within states.

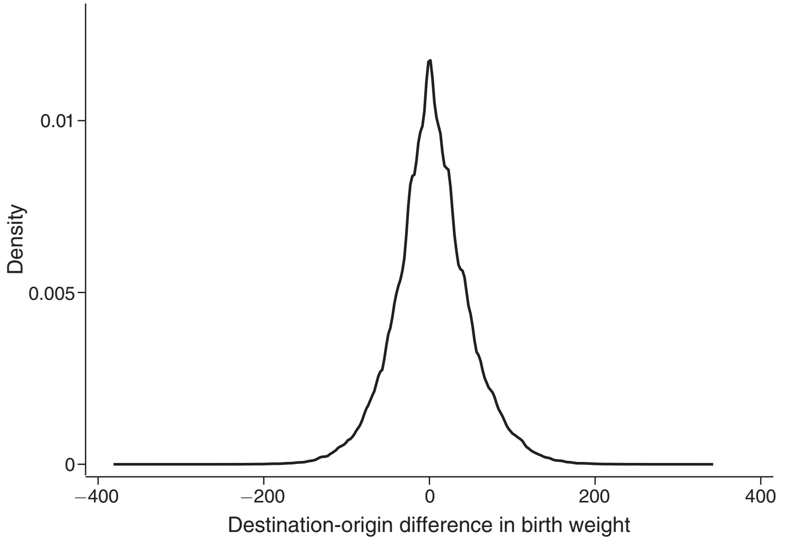

Figure 1. Distribution of Destination-Origin Difference in Average Child Birth Weight

Note: This figure shows the kernel density of the zip code–level destination-origin difference in average child birth weight ($\hat{\delta}_m$) for mothers who move across zip codes.

IV. Main Results

We begin our analysis by exploring event-study approaches that estimate the effects of moving to higher-quality areas. Next, we estimate place effects on birth weight at the zip code level and provide additional results at the city and county levels. Finally, we examine heterogeneity in place effects on birth weight, place effects on other important birth outcomes (summarized using difference-in-difference models), and the stability of place effects.[17]

A. *Initial Evidence on the Impact of Place*

As an initial exploration of the effect of moving to a new area, we plot the postmove change in birth weight for children of movers against the change in quality of a mother's location, $\hat{\delta}_m$. The slope of this graph can be interpreted as the extent to which moving to a location with higher average birth weight generates improvements in the early-life health outcomes of one's own children. If all geographic variation is due to the impact of place, we expect this plot to have a slope of 1. Alternatively, if individual factors explain all variation in birth weight, this plot should have a slope of 0.

[17] Specifically, we estimate the following equation: $y^c_{mjkt} = \alpha_m + \theta\hat{\delta}_m \times post_{mk} + \omega_k + \nu_t + \mathbf{x}'_{mk}\boldsymbol{\eta} + \varepsilon^c_{mjkt}$. Relative to our event-study model, we replace the relative birth-order dummies with $post_{mk}$, a single indicator for all births that occur after a mother's move. The key coefficient of interest is θ, the parameter on the interaction term $\hat{\delta}_m \times post_{mk}$.

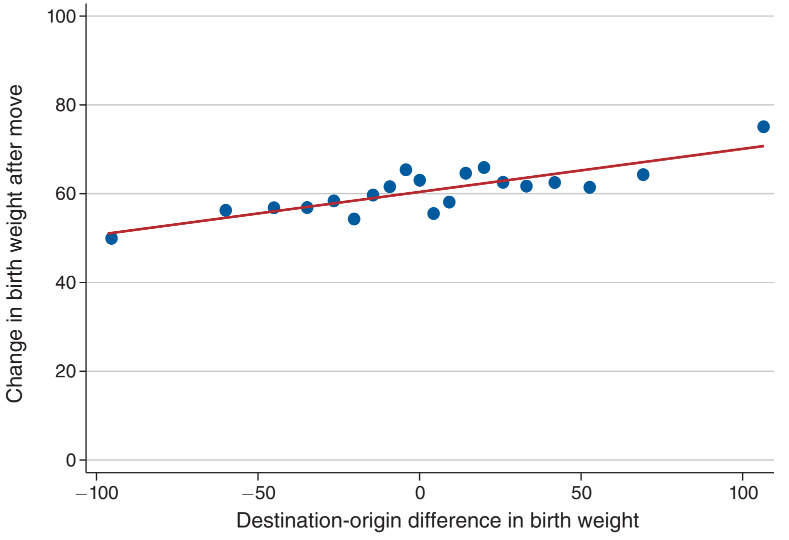

Figure 2. Destination-Origin Differences in Average Child Birth Weight

Notes: This figure shows the relationship between changes in child birth weight before and after a move across zip codes and the change in local-area birth weight that a mother experiences. For each mover, we calculate the difference $\hat{\delta}_m$ in average birth weight between their destination and origin zip codes and group the data into 20 bins. The x-axis displays the mean of $\hat{\delta}_m$. The y-axis reports binned averages of the change in birth weight for the children born before and after the move. The line of best fit is obtained from an OLS regression using the underlying mother-level sample.

The results in Figure 2 suggest that moving to a better location positively influences child birth weight. The slope of the fitted line is 0.10, which implies that place-based factors explain 10 percent of the geographic variation in birth weight. Notably, the pattern in the figure is symmetric above and below zero and appears to be linear. In particular, a move to a better or worse location is expected to generate similar absolute effects on birth weight. These findings indicate that mothers are prone to both the health costs and benefits of locations. They also support the assumption of a linear relationship between birth weight and $\hat{\delta}_m$ in equation (5).[18,19]

Next, Figure 3 reports event-study results by plotting the estimated coefficients on $\hat{\delta}_m$ from equation (5) for each birth relative to a move. For ease of presentation, we scale the coefficients to represent the impact of moving to a destination zip code that has a 100-gram-higher average birth weight than one's origin zip code (i.e., $\delta_m = 100$). The omitted category is the relative period $\theta_{r(m,k)} = -1$.

Our main estimates are plotted in solid gray triangle markers and show a statistically significant jump in weight for the first birth after a move. The magnitudes of the coefficients suggest that moving to a destination with a 100-gram-higher average birth weight leads to a 10-gram improvement in the weight of one's *own*

[18] It also supports the assumption of additive separability in equation (1), which requires that the change in outcomes when an individual moves from j to j' is the same as that when an individual moves from j' to j.

[19] In Figure 2, the change in birth weight is positive for all values of $\hat{\delta}_m$ (including when $\hat{\delta}_m = 0$) due to birth-order effects. We include fixed effects for birth order in our decomposition and event-study specifications to absorb this (level) effect.

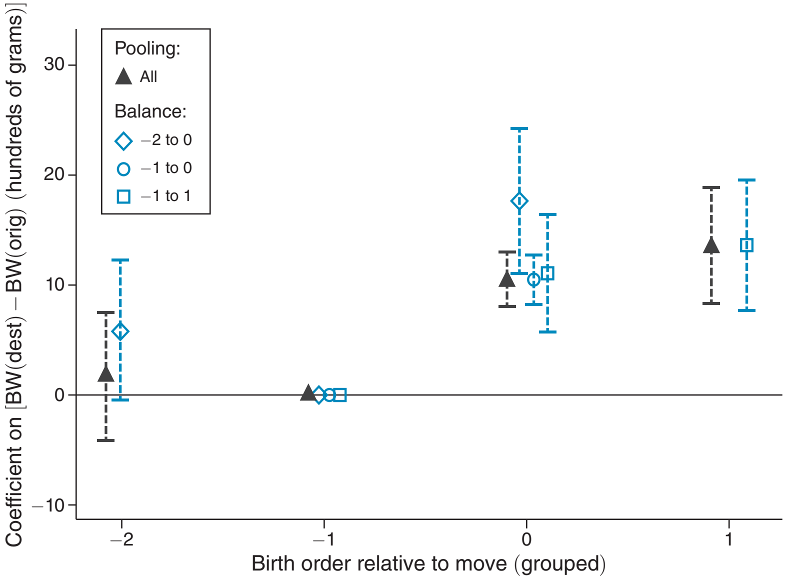

Figure 3. Event-Study Analysis of Child Birth Weight

Notes: This figure reports the coefficient estimates of $\hat{\theta}_{r(m,k)}$ from equation (5) in an analysis of zip code–level moves. The coefficient for the birth that occurs immediately before the move is normalized to 0. The x-axis indicates the birth order relative to the mother's move, $r(m,k)$. Each dot is a point estimate and represents the impact on birth weight measured in grams. The dashed vertical lines surrounding each dot are estimates of the 95 percent confidence interval. We report results for a sample that includes all mothers, as well as for three subsamples of mothers that are balanced in event time to estimate subsets of the event-study coefficients. The filled gray triangles report "pooled" estimates based on all mothers; the blue hollow diamonds report estimates based on the sample of mothers that have three births where two occur prior to a move and one occurs after a move; the blue hollow dots report estimates based on the sample of mothers that have two births where one occurs after a move; and the blue hollow squares report estimates based on the sample of mothers that have three births where one occurs prior to a move while two occur after a move. To reduce noise in the figure, the "pooled" coefficient shown at "−2" and "−1" on the x-axis includes the second and third birth prior to a move (i.e., $r(m,k) = -2$ and $r(m,k) = -3$) and the "pooled" coefficient shown at "1" on the x-axis includes the second and third births after a move (i.e., $r(m,k) = 1$ and $r(m,k) = 2$). Standard errors are clustered at the origin zip code level.

child, consistent with the slope of Figure 2. Moreover, the coefficient is essentially the same for the second birth after a move, indicating that the impact of a place is relatively constant over time.

A key estimate of interest in Figure 3 is the coefficient for the birth that occurs in relative period $r(m,k) = -2$, which constitutes our main test for the existence of pre-trends in birth weight. We do not find detectable evidence of differential trends. The point estimate is 1.7 and is statistically insignificant.

As a point to consider for the interpretation of our results, we note that the main event-study estimates rely on an unbalanced panel of mothers (i.e., we do not observe all mothers for two births before and two births after a move). This implies that the event time estimates are identified by distinct samples of mothers. We find similar estimates when we alternatively use balanced samples. The hollow blue markers in Figure 3 report results from separate event-study specifications where the samples are defined based on three different balance subsample restrictions.

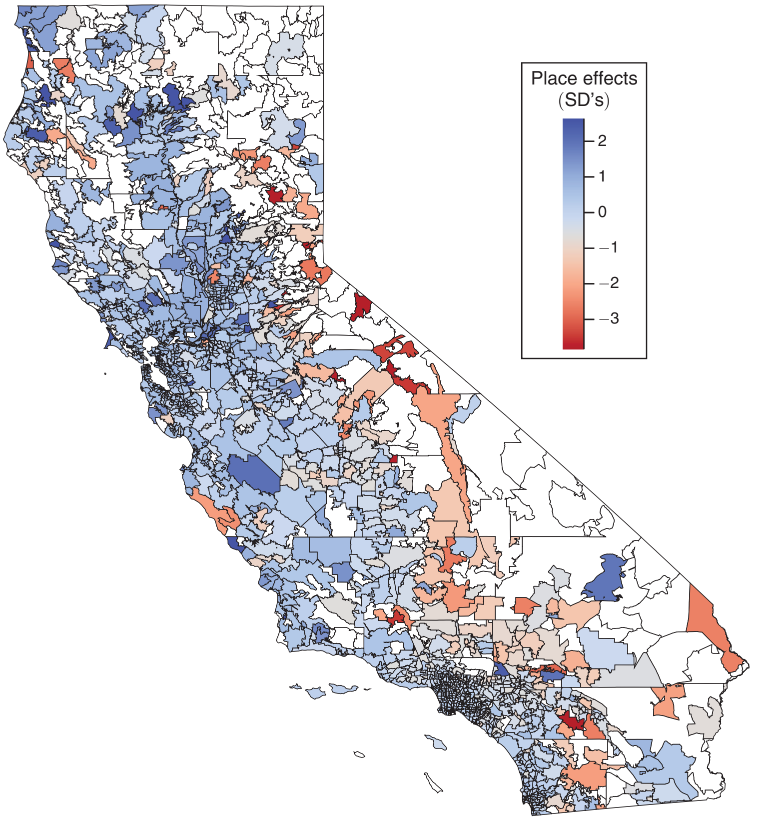

Figure 4. Estimated Place Effects

Notes: This figure provides a map of the estimated zip code place effects in standard deviation units. The legend indicates the level of birth weight with dark blue and dark red colors indicating zip codes with the highest and lowest estimated effects (relative to the omitted zip code, 92114), respectively. Zip codes with no color are not included in our estimation of zip code fixed effects. Note that the zip code effects are winsorized at the 1 percent level to limit the influence of outliers.

These include (i) mothers who have two births with one birth occurring after a move, (ii) mothers who have three births with two births occurring after a move, and (iii) mothers who have three births with two births occurring before a move. Reassuringly, we estimate a very similar increase in birth weight postmove as in the main estimates (in the gray triangle markers) and continue to find no detectable evidence of pre-trends.

B. *Impacts of Moving and Decomposition Results*

Next, we exploit the variation in birth location around a move to estimate equation (1). Figure 4 provides a map of the estimated $\hat{\gamma}_j$s. These estimates are relative to the effect of an excluded zip code (92114, in San Diego County) as we must omit one fixed effect due to collinearity. The estimates are roughly half-positive and half-negative, indicating that the place effect of the omitted zip code is close

Table 2—Estimated Impacts of Moving and Decomposition Results

	Zip code			City	County
	Top versus bottom 50% (1)	Top versus bottom 25% (2)	Top versus bottom 10% (3)	Top versus bottom 50% (4)	Top versus bottom 50% (5)
Panel A. Effects of moving					
Estimated impact (grams)	18.440	25.821	35.966	24.930	26.430
	(2.511)	(3.856)	(8.018)	(3.753)	(8.318)
Panel B. Decomposition					
Overall difference (grams)	108.886	179.448	263.573	104.932	65.299
Share due to place	0.169	0.144	0.136	0.238	0.405
	(0.023)	(0.021)	(0.030)	(0.036)	(0.127)
Share due to family (mother)	0.831	0.856	0.864	0.762	0.595

Notes: This table presents estimates of (i) the impact of moving to more advantaged areas on birth weight (panel A) and (ii) additive decomposition results of the role of place- and family-based factors in explaining the gap in birth weight across locations (panel B). All results are based on estimates of equation (1), where the dependent variable is birth weight (in grams). Columns 1–3 report results based on analysis of place effects at the zip code level, while columns 4 and 5 provide results for place effects at the city and county levels, respectively. The analysis is based on comparisons across sets of areas R and R'. For example, we define R and R' as the top 50 percent and bottom 50 percent of zip codes in terms of birth weight in column 1. The first row in panel A reports the estimated difference due to place effects (i.e., $\gamma_R - \gamma_{R'}$). The first row of panel B presents the overall difference in average birth weight between two areas (i.e., $\bar{y}_R - \bar{y}_{R'}$). The second and third rows of panel B report the shares of the overall difference attributable to place and family (mother) characteristics. Standard errors (in parentheses) are calculated using a mother-level bootstrap approach that has 50 repetitions.

to the median.[20] Suggestively, zip codes that are farther inland appear to have more negative place effects, although there are "good" and "bad" locations in most areas of the state. To contextualize the role of place in infant health, we now conduct two sets of analyses that describe the absolute and relative impact of location quality on infant birth weight.

Impacts of Moving to More Advantaged Areas.—First, we use our estimated zip code effects to estimate the absolute impact of moving to a higher-quality area. At baseline, we define this as the expected gain in birth weight associated with a move from a below- to an above-median-birth-weight zip code.[21] We also consider a range of alternative definitions of "higher-quality" areas, including comparisons of the top and bottom 25 and 10 percent of zip codes.

The first row of Table 2 shows that moving to an above-median zip code increases birth weight by 18 grams ($\text{SE} = 2.511$). This represents an 8 percent improvement relative to the average within-mother change in weight across births. It also compares favorably with the effects of policies that target maternal health. For example, it is similar to the impact of Medicaid eligibility during pregnancy, which has an 18-gram effect (East et al. 2021); it is within the range of impacts of participation

[20] We chose this zip code in part because it is relatively large (with 31,040 births and 3,349 movers) and diverse (294 White, 579 Black, 1,640 Hispanic movers). We convert the estimated place effects to standard deviations for easier interpretation.

[21] We estimate this as the difference between the average place effect for above- and below-median zip codes (i.e., $\hat{\gamma}_R - \hat{\gamma}_{R'}$).

in food stamp programs during pregnancy (Almond, Hoynes, and Schanzenbach 2011), which has a 15-to-42-gram effect; and it is comparable to reducing smoking during pregnancy by one cigarette per day, which has a 15-gram effect (Permutt and Hebel 1989).[22]

The remaining columns of Table 2 show the effects of moving when we consider more stark changes in the quality of a mother's location. The estimated effects in columns 2–3 indicate that the effect of moving scales up with larger changes in zip code quality. For example, moving from a bottom 10 percent to a top 10 percent location increases birth weight by 36 grams, which is twice as large as the effect of moving from a bottom 50 percent to a top 50 percent location.

Decomposition Analysis.—Second, we conduct two types of decomposition analysis of the overall gaps in birth weight between areas. As previewed in Section I, our main decomposition computes the share of the total difference in birth weight attributable to factors specific to place or family (mother) (i.e., $S_{place}(R,R') = (\gamma_R - \gamma_{R'})/(\bar{y}_R - \bar{y}_{R'})$ for groups R and R'). Rows 2–4 of Table 2 report results using the various definitions of relatively advantaged and disadvantaged areas.

Column 1 decomposes the difference in birth weight between above- and below-median zip codes. Overall, the gap is 109 grams, which is equivalent to roughly 4 percent of the average birth weight. The 18-gram causal increase in birth weight due to place effects represents 16.9 percent of the total difference. This implies that the remaining 83.1 percent is due to family (maternal) factors. The standard error on our estimate implies that we can reject a role for place that exceeds 21 percent or falls below 12 percent.

Columns 2 and 3 show similar but smaller shares attributable to place when we examine other definitions of high- and low-birth-weight locations. We consider a range of alternatives, including comparisons of the top and bottom 25 and 10 percent of zip codes. Moving across columns, the overall difference in average birth weight increases substantially, from 179 to 264 grams. Nevertheless, the estimated share of the gap explained by place is relatively stable, ranging from 13.6 to 14.4 percent. Hence, the majority of across-zip-code differences in birth weight appear to reflect *sorting* of mothers.

As an alternative to the additive decomposition, Table 3 decomposes the variance in birth weight across zip codes. The bottom of the table reports results for two scenarios: (i) the share of cross-zip-code variance in birth weight that would be eliminated if average maternal characteristics were equalized across zip codes and (ii) the share of cross-zip-code variance that would be eliminated if place fixed effects were equalized.

The variance decomposition shows that 79 percent of the variance in birth weight would be eliminated if maternal characteristics were equalized, while 5 percent of the variance would be eliminated if place effects were equalized. This is consistent with the results from the additive decomposition, which similarly suggested that place effects account for a minority of the disparity across locations. We also

[22] We obtain the impact of participation in Medicaid during pregnancy from East et al. (2021) by dividing the 1.2-gram increase in first-generation health (table A.3) by the 6.6 percentage point increase in eligibility (table 1).

TABLE 3—VARIANCE DECOMPOSITION OF GEOGRAPHIC DIFFERENCES IN BIRTH WEIGHT

	Estimates (1)
Variances of birth weight (grams)	
Birth weight	5,493.109
Place effects	1,183.564
Family (mother) effects	5,222.317
Corr. of average place and family effects	−0.184
	(0.139)
Share of variance reduced if:	
Family (mother) effects were made equal	0.785
	(0.102)
Place effects were made equal	0.049
	(0.118)

Notes: This table presents variance decomposition results for our zip code–level estimates of place effects. All results are based on estimates of equation (1), where the dependent variable is birth weight (in grams). We use a split-sample approach to estimate variances and covariances as detailed in Section II. The first row reports the variance of zip code average birth weight (i.e., $\bar{y}_j$). The second, third, and fourth rows report the variance of place effects (i.e., γ_j), the variance of family (mother) effects (i.e., $\bar{y}_j^m$), and the correlation of place effects and family effects. The final two rows report the shares of the variance in birth weight that would be reduced if zip code–level place effects were made equal and if family (mother) effects were made equal, respectively. Standard errors (in parentheses) are calculated using a mother-level bootstrap approach that has 50 repetitions.

find that there is a modest (but insignificant) negative correlation between $\bar{y}_j^m$ and γ_j, indicating that mothers with more advantaged characteristics (in terms of infant birth weight) tend to sort into areas that have slightly less beneficial place effects on child health.

C. *Place Effects at the City and County Levels*

While our main analysis focuses on place effects at the zip code level, a natural alternative is to consider the influence of place at more aggregate geographies such as cities and counties. One motivation for studying city and county effects is that local policies and programs are often determined within these boundaries, which could imply a stronger effect of moves across these lines. Moreover, prior research on place effects has often estimated place effects at higher geographic levels such as counties or commuting zones (CZs).[23]

Columns 4 and 5 of Table 2 show that moving to an above-median city or county increases birth weight by 25 (SE $=$ 3.7) and 26 (SE $=$ 8.3) grams, respectively. These effects are both substantively larger than the 18-gram increase that we find when studying moves across zip codes. In line with this, we also find that city- and county-level place effects account for a larger share of the cross-geography

[23] For example, Chetty and Hendren (2018) study causal exposure effects on upward mobility by studying children whose families moved at least 100 miles between CZs. We do not estimate effects at the CZ level because there are only 18 CZs covered by our data.

variation in birth weight. We find that place effects can account for 24 and 41 percent of the variation in birth weight across cities and counties, respectively.[24]

Overall, these results suggest that the impact of a move depends on the *type* of move as well as the change in location quality. We examine possible sources for the larger effects of moves across cities and counties and find the greatest support for an explanation based on the longer distance of these moves. To make this case, we first augment our event-study model and show that longer-distance *zip code–level moves*, relative to shorter-distance moves, have larger impacts on birth weight—nearly as large as our estimated effects of moving across counties (see Supplemental Appendix Table A3). Second, as detailed in Section VB, we find additional evidence for a distance-based explanation in that county-level moves generate the largest changes in the observable environmental inputs that are correlated with infant health.

How should we think about the magnitudes of these results? Relative to past estimates of the share of health-related outcomes explained by place, our estimates are at the lower end of the spectrum. At the higher end, place effects have been shown to account for at least 54 percent of the gap in seniors' health care utilization across hospital referral regions (Finkelstein, Gentzkow, and Williams 2016) and 70 percent of the gap in alcohol consumption across states (Hinnosaar and Liu 2022). On the lower end, Finkelstein, Gentzkow, and Williams (2021) find that 15 percent of the variance in elderly mortality across CZs would be eliminated by equalizing place effects. Our estimates suggest that the share of birth weight explained by place is less than the shares for health care utilization and alcohol purchases (at any geography we consider) and at least as large as the share for mortality. This is potentially consistent with an intuitive hypothesis that place is more influential for *flows* of health inputs than for *stocks* of health capital.

D. *Heterogeneity*

Next, we study whether the influence of place on infant health varies with socioeconomic background or race. This analysis is broadly motivated by previous research suggesting that contextual factors may have stronger impacts on disadvantaged populations in some settings (Currie and Walker 2011; Almond, Currie, and Duque 2018; Alexander and Schwandt 2022).

Table 4 presents the results for each level of geography, focusing on the impact of moving from a below- to above- and below-median area. Column 1 reproduces our baseline estimate from Table 2, and columns 2–5 report estimated place effects and decomposition results estimated separately for mothers who are college-educated, non-college-educated, White and non-Hispanic, and Hispanic, respectively. We study effects separately for Hispanic mothers because they constitute the predominant minority group in our sample.

[24] Supplemental Appendix Table A2 provides descriptive statistics for these samples. Supplemental Appendix Figure A3 presents the event studies for cross-city and cross-county moves and demonstrates that there are no detectable pre-trends.

TABLE 4—ESTIMATED IMPACTS OF MOVING AND DECOMPOSITION RESULTS, BY GROUP

	All (1)	College educated (2)	Non-college-educated (3)	White (4)	Hispanic (5)
Panel A. Zip code					
Effect of moving	18.440	7.101	21.976	28.432	22.275
	(2.511)	(4.083)	(3.287)	(2.839)	(3.839)
Share due to place	0.169	0.059	0.201	0.303	0.252
	(0.023)	(0.034)	(0.030)	(0.030)	(0.043)
Panel B. City					
Effect of moving	24.930	12.884	29.411	30.276	24.531
	(3.753)	(5.462)	(4.341)	(3.484)	(5.593)
Share due to place	0.238	0.111	0.281	0.310	0.275
	(0.036)	(0.047)	(0.042)	(0.036)	(0.063)
Panel C. County					
Effect of moving	26.430	24.998	40.136	43.772	44.132
	(8.318)	(11.787)	(6.856)	(5.384)	(9.653)
Share due to place	0.405	0.277	0.617	0.688	0.624
	(0.127)	(0.131)	(0.105)	(0.085)	(0.136)

Notes: This table presents estimates of (i) the impact of moving from a below- to an above-median-birth-weight area on a child's own birth weight and (ii) additive decomposition results of the role of place-based factors in explaining the share of the overall gap in birth weight between below- and above-median-birth-weight locations. Column 1 reproduces our pooled estimate from Table 2. Columns 2–5 report results for place effect estimates specific to the groups of college-educated, non-college-educated, White, and Hispanic mothers, respectively. All results are based on estimates of equation (1), where the dependent variable is birth weight (in grams). Panels A, B, and C provide results using place effect estimates at the zip code, city, and county levels, respectively. Standard errors (in parentheses) are calculated using a mother-level bootstrap approach that has 50 repetitions.

The most notable pattern across subgroups is that the importance of place is significantly larger and more precise for mothers with less than a college education. For mothers with a college education, the results in panel A show that moving to an above-median zip code leads to a marginally significant 7-gram improvement in birth weight, such that place explains roughly 6 percent of the birth weight gap (column 2). In contrast, moving to an above-median zip code increases birth weight by 22 grams for non-college-educated mothers, and place effects account for nearly 20 percent of the variation across place.[25,26] The patterns for city- and county-level moves in panels B and C are qualitatively similar, although the absolute effects of moves are larger for all subgroups.

The remaining results in the table suggest that the pattern by education does not appear to be driven by differences in the ethnic makeup of less educated mothers. In columns 4–5, we find a broadly similar role of place for both White and Hispanic

[25] Supplemental Appendix Figure A4 presents event studies estimated separately for college- and non-college-educated mothers (holding $\hat{\delta}_m$ constant). We do not find evidence of pre-trends for either group, and the impacts of moving are consistent with the decomposition. These patterns are unchanged when we instead define $\hat{\delta}_m$ using group-specific average birth weight.

[26] Given that non-college-educated mothers are slightly overrepresented among movers (see Table 1), this suggests that our main effects may be higher than the role of place in the population of California mothers. However, Supplemental Appendix Table A4 shows that reweighting our estimates to account for this imbalance between movers and nonmovers (as in Miller, Shenhav, and Grosz 2023) hardly changes our estimated effects.

mothers. Thus, it does not appear that racial background determines the sensitivity of women to the local environment.

E. *Stability of Place Effects*

A related question is how stable are place effects across groups and over time. We study this by correlating the estimated place effects across subgroups and across moves during two time periods: the first half of years (i.e., 1989–2004) or the second (i.e., 2005–2017).[27] Supplemental Appendix Table A5 shows that place effects at the zip code, city, and county levels are positively correlated across groups and time periods. College and noncollege place effects have a correlation between 0.35 and 0.70 across geographic levels. We find similar correlations between Hispanic and White place effects—a finding in contrast to evidence of small correlations of place effects across demographic groups in Israel (Aloni and Hadar 2023). We also find strong correlations in place effects over time ranging between 0.27 and 0.72.[28] This finding bears similarity to the stability of place effects on upward mobility (Chetty et al. 2020a).

F. *Impact of Place on Other Measures of Infant Health*

Finally, to better understand the biological pathways behind the increase in birth weight, we investigate the impacts of place on a broader set of health markers. For brevity, we summarize the effects of place using a difference-in-difference version of our event-study specification. Supplemental Appendix Table A6 shows the estimated effect of moving to a destination with a 100-gram-higher average weight on each of the outcomes of interest and across different geographies of moves. We find consistent improvements in log birth weight and gestational age: birth weight increases between 0.3 and 0.8 percent, and gestational age increases by 0.33 to 1.3 days. We also find reductions in low birth weight ($<$ 2,500 grams), very low birth weight ($<$ 1,500 grams), and being born premature ($<$ 37 weeks), although these effects are not always precisely estimated. As with the effects on birth weight, we find that these impacts tend to increase with longer-distance moves.[29] This suggests that higher-quality locations improve birth weight in part by increasing gestation, and that at least some of the rise in birth weight is attributable to a reduction in more rare and costly birth outcomes, such as being low birth weight.[30]

[27] We define the year of a move as the year of the first childbirth in a new location.

[28] Supplemental Appendix Figure A5 is a scatterplot of the time-period-specific estimated place effects.

[29] Supplemental Appendix Figures A6 and A7 show event studies for low birth weight and gestational age for all mothers and for non-college-educated mothers. The impacts of moving are larger for non-college-educated mothers for both outcomes. Supplemental Appendix Figure A8 shows that reestimating our event-study model with log birth weight as an outcome and $\hat{\delta}_m$ defined in logs produces very similar findings to our main effects.

[30] Examining impacts across the distribution of birth weight, we find the largest improvements in the likelihood of having a birth weight close to the median ($\approx$ 3,300 grams), with relatively smaller effects at the lower and higher tails. See Supplemental Appendix Figure A9.

V. Extensions

Having established that place has a direct influence over infant health, we now turn to two exercises that shed light on mechanisms and key confounds.

A. *Local Correlates of Place Effects*

Our primary evidence on mechanisms comes from an analysis of how local area characteristics (e.g., pollution) correlate with place effects. To identify local area correlates of place effects, we estimate the correlation of the estimated $\hat{\gamma}_j$ from equation (1) with proxies for five types of contextual factors. These include (i) demographic and economic measures, such as maternal education, the local area poverty share, and the Black population share; (ii) the violent crime rate, to capture community stress; (iii) proxies for access to general health and prenatal care, such as the number of physicians and obstetrician-gynecologists (ob-gyns) per capita; (iv) measures of adult health investments, including the rate of insurance and smoking among adults, and an estimate of place effects on elder mortality from Finkelstein, Gentzkow, and Williams (2021); and (v) environmental measures, including average temperature and the level of fine particulate matter ($PM_{2.5}$) and ozone. We standardize each measure to have a mean zero and standard deviation of 1. We examine these correlates at the zip code and city levels since we have relatively large samples for both of these geographies.[31] Supplemental Appendix B provides details on each measure and the underlying data sources.[32]

Figure 5 plots the bivariate correlation for each of these factors and the 95 percent confidence intervals.[33] The solid circle and hollow diamond markers represent correlations at the zip code and city levels, respectively. Across correlates considered in this analysis, the associations are notably similar at both levels of geography. Supplemental Appendix Figure A10 also shows that we find correlations with the same sign at the county level, although the magnitudes and confidence intervals are much larger.

The most striking result from this analysis is that the level of pollution in an area has a significant and large negative correlation with its estimated place effect. This includes particularly sizable correlations with ozone, which range from -0.33 to -0.32 at the zip code and city levels, respectively. These results are consistent

[31] Because we have fewer counties in the data, the estimated correlations at that level (reported in Supplemental Appendix Figure A10) are less precise.

[32] Note that the correlates we analyze are measured at different levels of geography. Demographic and economic measures are at the zip code level and are derived from the 2000 US decennial census (Manson et al. 2019). Violent crime rates are at the county level and are derived from annual arrest statistics from 1990 to 2015 (California Department of Justice n.d.). Proxies for access to care and health investments are at the county level and are derived from the Area Health Resource Files from 2010 (Health Resources and Services Administration 2019). Environmental measures are at the zip code level and are based on data from 2007 to 2009 (the earliest years available) from the CalEnviroScreen database (California Office of Environmental Health Hazard Assessment n.d.).

[33] We also explore correlations between our zip code–level estimates of mother effects (i.e., $\bar{y}_j^m$) and means for specific maternal characteristics. Supplemental Appendix Figure A11 shows that family (mother) effects on birth weight are moderately correlated with the share of non-White mothers. Correlations are smaller in magnitude for other characteristics such as maternal age and education. Supplemental Appendix Table A7 reports results from a decomposition exercise that suggests that observable maternal characteristics explain a small share of the difference in birth weight between above- and below-median zip codes.

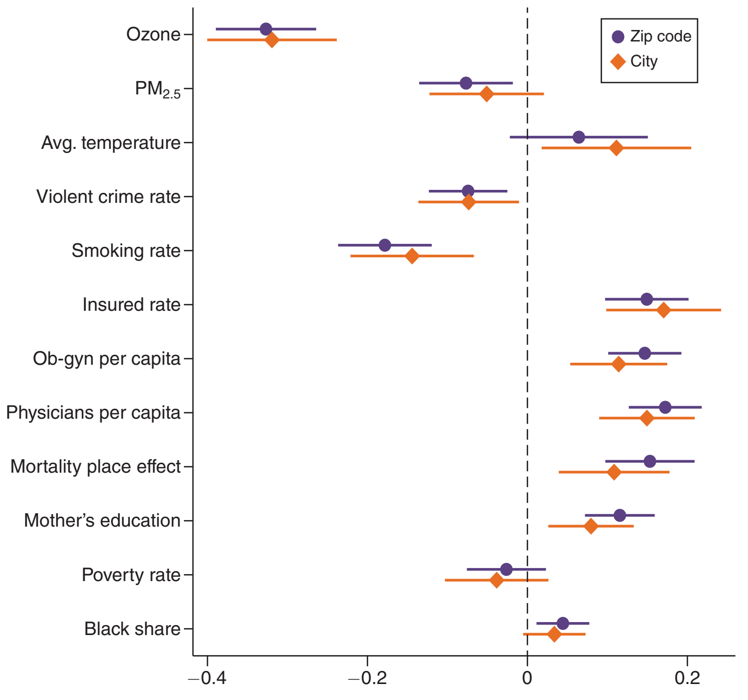

FIGURE 5. CORRELATES OF SPATIAL VARIATION IN ZIP CODE– AND CITY-LEVEL PLACE EFFECTS ON BIRTH WEIGHT

Notes: This figure shows the correlation of estimates of place effects based on equation (1) and place characteristics. For each characteristic listed on the *y*-axis, the markers report the point estimate of the correlation, and the horizontal lines show the 95 percent confidence intervals based on robust standard errors. The dot and diamond markers indicate whether the correlation is based on place effect estimates at the zip code or city level, respectively. Details on each measure and the underlying data sources are provided in Supplemental Appendix B.

with findings from experimental studies in animal models showing that exposure to ozone during pregnancy raises maternal pulmonary inflammation and reduces offspring weight (Gunnison and Hatch 1999). The results for particulate matter ($PM_{2.5}$) are smaller but nevertheless are statistically significant. These results broadly align with evidence that reductions in pollutants, including ozone and $PM_{2.5}$, improve infant weight (Alexander and Schwandt 2022). However, a back-of-the-envelope calculation suggests that other mechanisms likely also contribute to our estimated place effects.[34]

We also find smaller but still meaningful negative associations with other area characteristics. For instance, zip codes and cities that have higher violent crime rates have less beneficial place effects (around $\rho = -0.07$ for both geographies).

[34] To come to this conclusion, we rely on estimates from Alexander and Schwandt (2022), who study the effects of car pollution and find that a 2.6 percent increase in $PM_{2.5}$ (table 2, panel B of their paper) causes a 6.2-gram decrease in birth weight (table 3 of their paper). Based on these results, we expect that a move from a below- to an above-median zip code, which has a 15 percent difference in $PM_{2.5}$, would translate into a 36-gram increase in birth weight ($= 15 \times (6.2/2.6)$). This is roughly twice the impact that we estimate for this type of move, which suggests that $PM_{2.5}$ may be correlated with other place-based factors that offset the expected improvement from this channel.

Similarly, areas with a higher share of adults who smoke tend to have lower birth weight (with $\rho = -0.18$ and $\rho = -0.14$ for zip codes and cities). This is potentially consistent with experimental evidence that suggests that smoking has important impacts on birth weight (Permutt and Hebel 1989).

In terms of positive associations, we find important roles for the number of physicians, the causal effects of place on adult mortality, and ob-gyns per capita. At both geographies that we consider, these are among the largest correlations that we find and consistently exceed 0.10. In addition, we find a positive and significant correlation for the share of adults who are insured. These results highlight the potential importance of health care access.[35]

The remaining demographic, economic, and temperature measures tend to have smaller correlations.[36] Of these, the largest magnitude coefficient is for maternal education (the share of mothers with a college degree). This may reflect the fact that higher-educated women have higher levels of civic engagement and greater support for progressive policies (Milligan, Moretti, and Oreopoulos 2004; Gillion, Ladd, and Meredith 2020).

Finally, Figure 6 expands on these results by examining whether these correlations vary between college-educated and non-college-educated mothers. For both groups, we find that ozone has among the largest negative associations with place effects. However, the correlation with ozone is more than four times as large for mothers without college than for college graduates. We can statistically reject that the correlation is the same at both geographies (p-value < 0.01). This could suggest that less educated mothers are either more exposed to airborne pollutants or more susceptible to harmful effects from exposure. For the remaining characteristics, we often find similar correlations and largely overlapping confidence intervals between the results for higher- and lower-educated mothers.

B. *Changes in Maternal Behaviors around a Move*

Next, we test for changes in time-varying maternal circumstances and behavior around the time of a move. This could matter in two distinct ways for the interpretation of our results. First, even though we find no evidence of pre-trends in our event study, it remains possible that mothers who experience positive shocks between births could respond by moving to areas that have better place effects on child health. Second, mothers may respond to the experience of living in a higher-quality location by changing their behaviors, such as by starting a relationship with a higher-earning partner or securing a job that provides health insurance. In either of these cases, our estimated place effects could partly reflect confounds due to time-varying shocks or

[35] Supplemental Appendix Figure A12 presents correlations between place effects and other local characteristics, including a standard measure of life expectancy, upward mobility, and average birth weight. We find a strong positive correlation with mean birth weight, consistent with these causal effects explaining a portion of the gap between low- and high-birth-weight areas. We find little correlation with life expectancy or upward mobility.

[36] While we find smaller correlations for factors outside of pollution and ob-gyn access, many of these correlations are larger than those used to explain variation in health-spending place effects in Finkelstein, Gentzkow, and Williams (2016) (which are typically below 0.05).

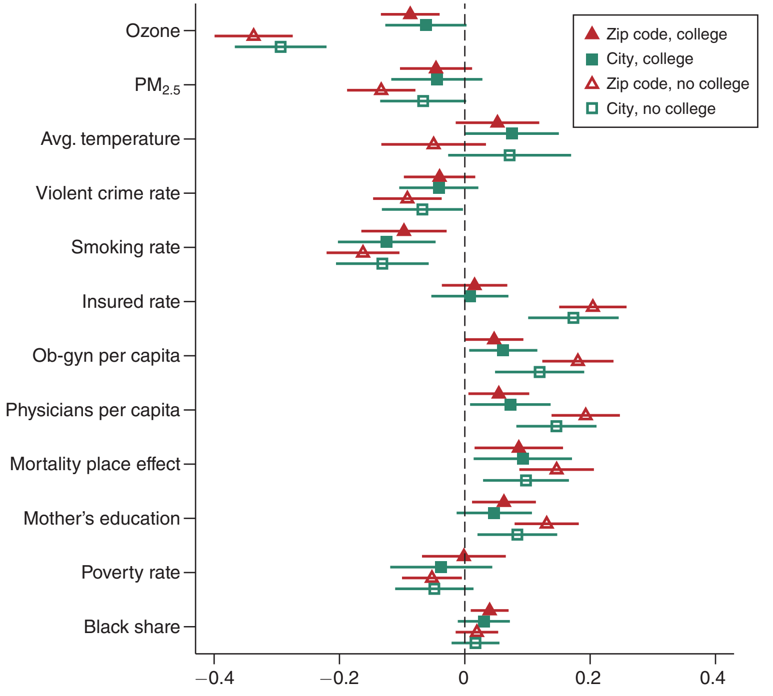

Figure 6. Correlates of Spatial Variation in Place Effects on Birth Weight, by Maternal Education

Notes: This figure shows the correlation of estimates of place effects based on equation (1), estimated separately for college-educated and non-college-educated mothers, and place characteristics. For each characteristic listed on the *y*-axis, the unfilled and filled markers report the point estimates of the correlations for non-college-educated and college-educated mothers, respectively, and the horizontal lines show the 95 percent confidence intervals based on robust standard errors. Triangle and square markers denote whether the correlation is based on place effect estimates at the zip code or city levels, respectively. Details on each measure and the underlying data sources are provided in Supplemental Appendix B.

the additional mechanism of changes in maternal inputs driven by moves to better areas, respectively.

To assess the potential importance of time-varying maternal factors, we perform an exercise in which we use family-level characteristics to try to predict the effects of moving on infant health. Specifically, we construct a maternal index for each infant as the fitted value from a regression of birth weight on a number of proxies for partner quality and financial resources.[37] This indexing approach allows us to aggregate many maternal characteristics into a single measure (which can provide more power to detect an effect), to assign greater weight to outcomes that are more correlated with birth weight, and to compare magnitudes with our main effects.

[37] These covariates include an indicator for whether a father is present at the time of birth, father's age, indicators for whether the father completed high school and college, and indicators for whether a mother has public or private insurance. Missing-father characteristics are imputed as the mean in a given calendar year. Based on the R^2 of the prediction regression, these covariates are one-third as predictive of birth weight as indicators for maternal race and education categories.

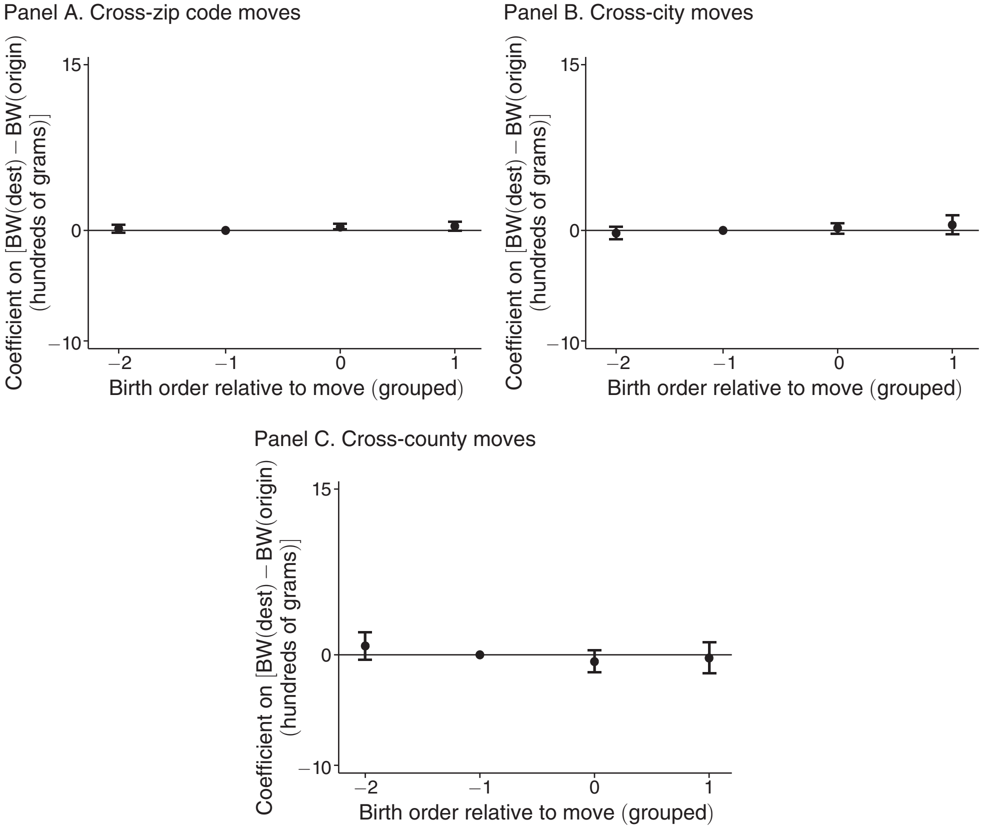

Figure 7. Predicted Infant Birth Weight (Maternal Index) around a Move

Notes: This figure shows coefficients from our event-study model (equation (5)) in which the dependent variable is *predicted* infant birth weight using maternal and household covariates. We show results separately for moves across zip codes (panel A), moves across cities (panel B), and moves across counties (panel C). We generate the predicted "maternal index" as the fitted value from a regression of birth weight on an indicator for whether a father is present at the time of birth, father's age, indicators for whether the father completed high school and college, and indicators for having public or private insurance. Missing-father characteristics are imputed as the mean in a given calendar year. See the notes for Figure 3 for additional details about the interpretation of the estimates.

Figure 7 presents event-study analyses showing the impact of a move to a 100-gram-higher-quality location on the maternal index. Overall, we find small and often insignificant effects across the various types of moves. For cross-zip-code moves (panel A), we find that the postmove increase in the maternal index is around 0.3 grams, which implies that these factors could at most explain 3 percent of our estimated impact of moving. For cross-city and cross-county moves (panels B and C), we find even smaller effects, between −0.7 grams and 0.3 grams, both of which are statistically insignificant. We also find insignificant effects on the majority of the individual components of the maternal index (although we find some increases in private health insurance, in line with the positive correlation with local insurance rates in Figure 5); see Supplemental Appendix Table A8. This provides suggestive evidence that the observed change in birth weight after a move is unlikely due to time-varying individual factors.[38]

To validate that the small effects on the maternal index are not due to lack of statistical power or poor model fit, we also examine the impact of moving on a second index that is based on place-based covariates. In particular, we generate a place-based index as the fitted value from a regression of birth weight on 11 of the 12 place-based characteristics in our correlational analysis.[39] Supplemental Appendix Figure A13 shows that the place-based index increases substantially after a move—regardless of the geography of the move. Importantly, this is not due to greater predictive power of place-based factors for infant health generally, as the R^2 of the prediction regression is in fact larger when we use maternal covariates. The increase in the index is two orders of magnitude larger than the effect that we predicted using maternal covariates. Consistent with longer-distance moves leading to a larger change in inputs, we find that the place-based index increases by 31 grams after a zip code–level move and by 45 grams after a county-level move.

In line with a minor role for maternal behaviors in our results, we find that our estimates are not sensitive to controlling for maternal circumstances around a move. Supplemental Appendix Table A9 shows that we find similar effects of moving when we control for the inputs into the maternal index or include couple-specific fixed effects.

VI. Robustness Exercises

In this section, we report results from several exercises that provide evidence in support of our key identifying assumption and assess the sensitivity of our results.

A. *Endogenous Location Decisions*

A primary threat to identification is that the quality of a mother's destination may be correlated with unobserved, time-varying determinants of child health. To address this type of concern, we leverage two sources of plausibly exogenous changes ("shocks") to location quality. As a first shock to location quality, we use the average change in location quality for all *other* mover mothers from a mother's origin as an instrument for the difference in average birth weight between her origin and destination (i.e., $\hat{\delta}_m$). Broadly similar approaches appear in other recent papers. For example, Chetty and Hendren (2018) reproduce their main exposure effect estimates by predicting the change in location quality using other movers from a child's origin. Relative to their approach, we innovate by combining an IV approach

[38] We find similarly small effects when we expand the index to include indicators for whether the mother worked in the last year, her expected income based on her reported occupation, and the average income for women by occupation calculated from the 2007–2017 American Community Surveys, whether she received any Special Supplemental Nutrition Program for Women, Infants, and Children (WIC) assistance for the pregnancy, and the average number of cigarettes smoked during pregnancy. These measures are only available after 2007, and we limit this robustness analysis to 2007 and onward. See Supplemental Appendix Figure A14. We separately study effects on WIC participation in Supplemental Appendix Figure A15 and do not find significant effects.

[39] We leave out the mortality place effect because it may be mechanically related to improvements in birth weight.

with an event-study framework. This uniquely allows us to examine the validity of the instrument by testing whether predicted changes in location quality affect birth weight for children born before a mother moves. This potentially reduces the scope for selection bias by eliminating variation in the change in location quality due to individual choice. Supplemental Appendix Table A10 shows that the instrumented point estimates are very similar to our main results.[40]

As a second shock to location quality, we attempt to study moves among families who are attached to the military. Specifically, we focus on individuals who ever live in or near a zip code with a high concentration of military personnel. We identify a military location as a zip code where a large number (5 percent) of parents report an occupation related to the military, such as "soldier."[41] Because military personnel often have less choice over location, the change in location quality due to assignment to or from their posting is less likely to reflect one's own personal circumstances. Interestingly, Supplemental Appendix Table A10 shows that these moves lead to *larger* improvements in child health than the average move. These larger treatment effects may reflect the longer-distance nature of these moves: Military moves are almost twice as likely as other moves to be across counties (57 percent versus 32 percent).

B. *Endogenous Fertility Decisions*

We also conduct an analysis to address the potential concern that fertility is correlated with a mother's location choice. For example, if mothers are less likely to have a birth when they move to a lower-quality location (smaller $\hat{\hat{\delta}}_m$), then the estimated postmove changes in birth weight could confound such differences in unobserved selection into having a birth across locations with the causal impacts of place. We examine these types of fertility responses by testing whether the quality of a mother's destination, $\hat{\hat{\delta}}_m$, is a significant predictor of a total completed fertility, controlling for fixed maternal characteristics.[42] Supplemental Appendix Table A11 shows that $\hat{\hat{\delta}}_m$ is not associated with completed fertility once minimal controls for maternal characteristics are included.[43] This provides strong evidence that selective fertility is not a factor in our estimates.

C. *Additional Sensitivity Analysis*

Finally, we examine whether our results are sensitive to measuring a mother's location quality using only nonmovers or to changes in our sample criteria.

[40] Notably, when we run the event study, the IV estimate for the coefficient for the birth that occurs in relative period $\theta_{r(m,k)} = -2$ is not statistically significant; see Supplemental Appendix Figure A16. This favors the plausibility of the exclusion restriction in our setting.

[41] This is the top 1 percent of military concentrations across zip codes. It is common for parents in these locations to have their occupation withheld in the birth records, which we view as consistent with an association with the military. We identify military locations using the 2007–2017 birth records, where parent occupations are available, and apply the geographic-specific definition to all mothers in our sample (regardless of when they give birth).

[42] Controls include fixed effects for mother race, whether a mother has a college education, and the age of a mother's first birth.

[43] The controls include fixed effects for maternal race, completed education, and age at first birth.

Supplemental Appendix Figure A17 shows the event-study results when we define $\hat{\delta}_m$ using only nonmovers, so there is no overlap between the mothers used to identify the treatment effect of moves and the mothers used to estimate $\hat{\delta}_m$. The patterns are very similar to our main effects. Supplemental Appendix Tables A12 and A13 demonstrate that our decomposition results are not meaningfully changed when we expand our sample to include mothers that move multiple times or limit it to locations with a larger number of movers (at least 100) to reduce possible measurement error in our estimates.

VII. Discussion

One final question is the extent to which being born in a better location is likely to improve *long-run* outcomes. Understanding the size of any lasting effects is relevant for gauging whether improved birth outcomes could be a mechanism for previously documented impacts of childhood neighborhoods on intergenerational mobility, as well as for inferring the size of the potential fiscal externality from these moves.

Because we are unable to observe long-run outcomes directly, we project our place impacts on birth weight to achievement and earnings using estimates of the effect of birth weight from previous studies based on twin comparisons (Black, Devereux, and Salvanes 2007; Figlio et al. 2014). This is likely to be a conservative estimate of the total effect on outcomes since improved location can affect fetal development in ways not captured by birth weight (see, e.g., Persico, Figlio, and Roth 2020). Nevertheless, the estimated 18-gram (0.56 percent) gain in birth weight from moving between a below- and above-median zip code would be expected to lead to 0.31 percent of a standard deviation improvement in test scores and a 0.06 percent increase in earnings. Moreover, for longer-distance moves, these effects are projected to be roughly 40 percent larger, based on the estimated 26-gram (0.8 percent) gain in birth weight from moving between a below- and an above-median county (i.e., a 0.42 percent of a standard deviation improvement in test scores and a 0.08 percent increase in earnings).

Taking the long-run projections at face value, these results suggest that in utero exposure to better neighborhoods likely explains a small share of how childhood neighborhoods shape adult outcomes. This is consistent with evidence that the majority of prior place-based benefits appears to accrue during adolescence, with smaller impacts during early childhood (Deutscher 2020; Chetty et al. 2020b). Collectively, this indicates that childhood place effects on long-run outcomes operate primarily through postbirth mechanisms, such as schools or neighborhood peer effects.

VIII. Conclusion

This paper uses birth records and a movers-based research design to estimate the absolute and relative importance of place effects for early-life health. We find that the gain in birth weight from moving to a higher-quality location (as proxied by higher average birth weight) is comparable to the effects of policies directly targeting maternal health, even for very local (zip code–level) moves. We find that place effects are larger for longer-distance moves and more influential for children of non-college-educated mothers. A descriptive analysis suggests that causal place effects are most related to the presence of airborne pollutants, particularly the level of ozone.

Overall, our analysis provides new evidence on the total role of place-based factors on infant health. The estimates capture the effects of more-difficult-to-measure characteristics of neighborhoods, such as the degree of social capital, and more salient features studied in prior work, such as pollution and climate. Our work provides new evidence about the importance of neighborhoods for infant development, complementing recent research that has found large impacts of childhood location on long-run outcomes (Chetty, Hendren, and Katz 2016; Chetty et al. 2020a). Based on our estimates, a back-of-the-envelope calculation suggests that a small portion of the lasting effects of neighborhoods could be due to improvements in early-life health.

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